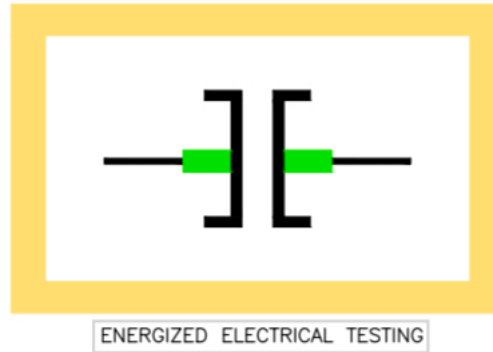




ELEC2461 – TEST REPORT

Yuvraj Grewal & Jacob

Energized Electrical Testing

Submitted to:

Mr. Yang Shu

December 8, 2025

NAIT

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Letter of Transmittal

Mr. Yang Shu

Northern Alberta Institute of Technology

11762-106 St. NW Edmonton, AB, Canada, T5G 2R1

We're pleased to inform you that the testing of the spare field equipment you provided has been completed. Below is a list of all equipment we tested:

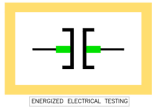
1. Westinghouse CO Relay
2. GE-735 Relay
3. GE-750 Relay
4. Basler-951 Relay
5. SEL-751 Relay
6. SEL-587 Relay
7. GE-469 Relay

These tests were carried out between September and December 2025. This report also includes equipment descriptions; and a summary of deficiencies found between tests. For detailed test sheets, please refer to Section 4.

If you have any feedback or questions, feel free to reach out to us using the contact information provided below. We sincerely appreciate the opportunity to assist with your field equipment testing and hope to work with you again in the future.

Warm Regards,

Jacob Dardon & Yuvraj Grewal

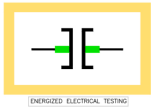


1. Summary

This report documents the inspection and electrical testing of seven protective relays provided by NAIT, following NETA MTS, manufacturer, and NAIT standards. Testing was performed between September and December 2025. All relays went through visual and mechanical inspection and electrical testing for the protection elements, including phase/ground overcurrent, differential, directional elements, overload, RTD biasing, and motor protection features, depending on the relay type.

Most relays successfully met the required pickup, timing, metering, directional, and differential test. However, several relays showed deficiencies that needs attention. The Westinghouse CO relay showed proper electrical operation but had damaged mechanical flag mechanisms that requires replacement to maintain accurate trip indication. The GE-735 relay had the most significant issue, failing all 51 timing tests with errors well beyond acceptable limit. The relay should be removed from service until the cause is identified. The GE-469 motor protection relay performed adequately in most areas but failed the undercurrent alarm timing requirement and was near the error limit provided by the manufacturer, suggesting replacement may be necessary. For all relays that passed, routine maintenance such as cleaning, tightening connections, and periodic inspection was recommended to ensure long-term accuracy and operational longevity.

Overall, while the protection systems remain largely functional, the identified deficiencies highlight the importance of ongoing maintenance.



2. Scope of Work

Provided below is a list of all spare equipment provided by Northern Alberta Institute of Technology. Each equipment is followed by the applicable tests conducted with compliance to NETA MTS, Manufacturer's and NAIT standards.

1. Westinghouse CO relay

- Visual and Mechanical Inspection
- Electrical Tests
 - 51 Pickup and Timing
 - DC Seal-In
 - 50 Pickup and Timing

2. GE-735 relay

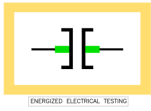
- Visual and Mechanical Inspection
- Electrical Tests
 - 51P Pickup and Timing
 - 50P Pickup and Timing

3. GE-750 relay

- Visual and Mechanical Inspection
- Electrical Tests
 - 51P Pickup and Timing
 - 50P Pickup and Timing

4. Basler-951 relay

- Visual and Mechanical Inspection
- Electrical Tests
 - 51P Pickup and Timing
 - 50P Pickup and Timing
 - 67P and 67N Instantaneous Pickup



5. SEL-751 relay

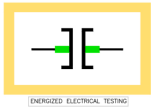
- Visual and Mechanical Inspection
- Electrical Tests
 - 51 Phase Pickup and Timing
 - 50 Phase Pickup and Timing

6. SEL-587 relay

- Visual and Mechanical Inspection
- Electrical Tests
 - Restrained Operating Current Pickup
 - Restraint Slope 1 Percentage
 - Restraint Slope 2 Percentage
 - Unrestrained Operating Current Pickup
 - Harmonic Blocking

7. GE-469 Relay

- Visual and Mechanical Inspection
- Electrical Tests
 - Overload (with/without RTD biasing, with/without current unbalance biasing)
 - Current Unbalance
 - Undercurrent
 - Mechanical Jam
 - Ground Fault
 - Phase Reversal



3. Tested Equipment

1. Westinghouse CO relay

The Westinghouse CO is an electromechanical inverse-time overcurrent relay. The 51-element uses a disk and magnet operating mechanism to sense and trip on excessive current with a time delay. It also provides a 50-element trip and a DC-seal-in.

2. GE-735 relay

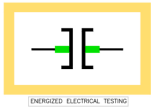
It is a microprocessor-based relay used to perform primary circuit protection. The relay features instantaneous, time overcurrent and ground protection trips. The relay also features “Trip Record” showing the last five causes of trips.

3. GE-750 relay

The GE-750 relay is a microprocessor-based protection and control device designed for primary and backup protection of buses, transformers, and. The relay also provides monitoring, event recording, and flexible programmable features, all configurable through the front panel or EnerVista software.

4. Basler BE1-951 relay

The BE1-951 relay is a multifunction microprocessor-based relay providing directional phase, ground, and negative-sequence overcurrent protection along with power, voltage, frequency, reclosing, and breaker-failure functions. It includes flexible programmable I/O and logic, which allows to assign outputs using BESTlogic.



5. SEL-751 relay

The SEL-751 relay includes combination of protection and high impedance fault location features. The relay also provides monitoring, control and communication in one package. The base relay includes voltage, current, frequency and power protection elements. It also comes with a interactive HMI touch-screen.

6. SEL-587 relay

The SEL-587 Relay is a current differential and overcurrent relay designed for protecting two-winding transformers, reactors, generators, and other two-terminal equipment. It measures high- and low-side currents, harmonics, and differential. The relay offers dual-slope differential elements for security during CT saturation and supports wye- or delta-connected CT configurations.

7. GE-469 relay

The GE SR469 Motor Protection System is a draw-out relay designed for medium voltage motors, offering advanced protection with customizable overload curves and single CT differential protection. It features simplified configuration through the Motor Settings Auto-Configurator and supports multiple communication protocols for seamless integration into control networks. This combination delivers flexible, cost-effective motor protection, control, and diagnostics.

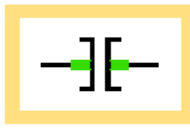
1. Test Data

4.1 Westinghouse CO Relay

Relay Inspection			
	Phase A	Notes	
Manufacture	Westinghouse		
Model	C08-D411BA1		
Serial Number	391B690A09		
Physical Condition	OK	There were no visible signs wear and tear	
50 Flag	Not OK	See deficiencies (Section 5)	
51 Flag	Not OK	See deficiencies (Section 5)	
Test Equipment			
Manufacture	MANTA	Serial Number	13599
Model	MTS - 5100	Calibration Date	April 30, 2017

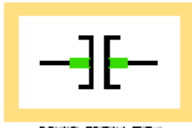
51 (Time Overcurrent) Phase Settings				
Set Points		Value		Tolerance
Pickup		2.5		3%
Timing Curve		CO-8		N/A
Time Dial		1		5%
51 Phase Pickup Test				
Setting	Calculated Value	Phase A		
		Result	Error	
2.5A	2.5A	2.464	1.44	
Comments:	PASS			
51 Phase Timing Test				
Multiple of PU	Injected Current	Calculated Trip Time	Phase A	
			Result	Error
2x PU	5	2.002	1.981	1.06%
3x PU	7.5	0.836	0.9390	1.19%
4x PU	10	0.508	0.502	1.18%
Comments:	PASS			

6

	Company: Energized Electrical	Work Order: Northern Alberta Institute of Technology
	Location: Edmonton, AB	Date: 12/8/25
	Tested by: Dardon.J, Grewal.Y	

50 (Instantaneous Overcurrent) Phase Settings			
Set Points		Value	Tolerance
Pickup		10A	10%
Time Delay		0	<50ms
50 Phase Pickup Test			
Setting	Calculated Value	Phase A	
		Result	Error
10A	10A	10.3A	3%
Comments:	PASS		
50 Phase Timing Test			
Setting	Injected Current	Phase A	
		Result	Pass / Fail
10A	12A	17.6ms	PASS
Comments:	PASS		

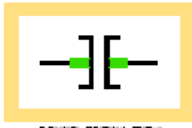
ICS (Indicating Contactor Switch) Test		
Setting	Phase A	
	Result	Pass / Fail
0.2A	0.2A	PASS
Comments:	PASS	

	Company: Energized Electrical	Work Order: Northern Alberta Institute of Technology
	Location: Edmonton, AB	Date: 12/8/25
	Tested by: Dardon.J, Grewal.Y	

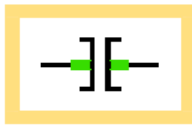
4.2 GE-735 Relay

5 Inspection		
	OK / N-OK	Comments
Cleanliness Of Relay	OK	No dust was accumulated on the relay
Physical Condition	OK	No visible signs of wear and tear
HMI Operation	OK	HMI was operational
LED Operation	OK	All LED were working correctly
Relay Serial Number	735-5-5-HI-485	
Test Equipment		
Manufacture	Manta	
Model	MTS-5100	
Serial Number	13599	
Calibration Date	April 30, 2017	

51 (Time Overcurrent) Phase Settings								
Set Points		Value		Tolerance				
Pickup		40%		3%				
Timing Curve		VI		N/A				
Time Dial		2		3%				
Shift Factor		0.8		N/A				
51 Phase Pickup Test								
Setting	Calculated Value	Phase A		Phase B		Phase C		
		Result	Error	Result	Error	Result	Error	
2A	2	2.008A	0.4%	2.013A	0.65%	2.002A	0.1%	
Comments:	PASS							
51 Phase Timing Test								
Multiple of PU	Injected Current	Calculated Trip Time	Phase A		Phase B		Phase C	
			Result	Error	Result	Error	Result	Error
2x PU	4	2.12	2.619	23.5%	2.66	25.5%	2.65	25%
3x PU	6	0.859	1.0665	24.2%	1.07	24.6%	1.07	24.6%
4x PU	8	0.546	0.6822	24.9%	0.681	24.7%	0.68	24.5%
Comments:	FAIL							

	Company: Energized Electrical	Work Order: Northern Alberta Institute of Technology
	Location: Edmonton, AB	Date: 12/8/25
	Tested by: Dardon.J, Grewal.Y	

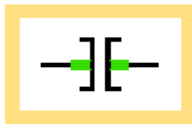
50 (Instantaneous Overcurrent) Phase Settings							
Set Points		Value			Tolerance		
Pickup		4			3%		
Time Delay		0			<35ms		
50 Phase Pickup Test							
Setting	Calculated Value	Phase A		Phase B		Phase C	
		Result	Error	Result	Error	Result	Error
4	20A	19.9A	0.5%	20.1A	0.5R%	20	0%
Comments:	PASS						
50 Timing Test							
Setting	Injected Current (A)	Phase A		Phase B		Phase C	
		Result(ms)	Pass or Fail	Result(ms)	Pass or Fail	Result(ms)	Pass or Fail
4	30	20	PASS	24.9	PASS	22.6	PASS
Comments:	PASS						

	Company: Energized Electrical	Work Order: Northern Alberta Institute of Technology
	Location: Edmonton, AB	Date: 12/8/25
	Tested by: Dardon.J, Grewal.Y	

4.3 GE-750 Relay

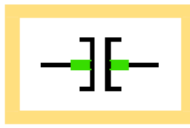
Inspection		
	OK / N-OK	Comments
Cleanliness Of Relay	OK	No dust was accumulated on the relay
Physical Condition	OK	No visible signs of wear and tear
HMI Operation	OK	HMI was operational
LED Operation	OK	All LED were working correctly
Relay Serial Number	E00962769	
Test Equipment		
Manufacture	Manta	
Model	MTS-5100	
Serial Number	13599	
Calibration Date	April 30, 2017	

Metering Tests								
Voltage Magnitude & Angle Tests								
PT Ratio		120:1			PT Configuration		WYE	
Injected Value			Calculated Reading			Actual Reading		
A	B	C	A	B	C	A	B	C
10	10	10	1.2KV	1.2kV	1.2kV	1.2kV	1.2kV	1.2Kv
Comments:		PASS						
Current Magnitude & Angle Tests								
CT Ratio		500:5			CT Configuration		WYE	
Injected Value			Calculated Reading			Actual Reading		
A	B	C	A	B	C	A	B	C
2	2	2	200	200	200	199	200	199
Comments:		PASS						

	Company: Energized Electrical	Work Order: Northern Alberta Institute of Technology
	Location: Edmonton, AB	Date: 12/8/25
	Tested by: Dardon.J, Grewal.Y	

51 (Time Overcurrent) Phase Settings								
Set Points		Value			Tolerance			
Pickup		2			0.5			
Timing Curve		VI			N/A			
Time Dial		2			0.3			
51 Phase Pickup Test								
Setting	Calculated Value	Phase A		Phase B		Phase C		
		Result	Error	Result	Error	Result	Error	
40%	2	1.998	0.1%	2.011	0.55%	2	0%	
Comments:	PASS							
51 Phase Timing Test								
Multiple of PU	Injected Current	Calculated Trip Time	Phase A		Phase B		Phase C	
			Result	Error	Result	Error	Result	Error
2x PU	4	2.624	2.66	1.4%	2.69	2.5%	2.64	0.6%
3x PU	6	1.075	1.08	0.5%	1.07	0.5%	1.09	1.4%
4x PU	8	0.687	0.683	0.6%	0.673	2%	0.668	2.8%
Comments:	PASS							

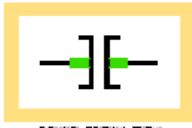
50 (Instantaneous Overcurrent) Phase Settings							
Set Points		Value			Tolerance		
Pickup		4			45ms		
Time Delay		0			0		
50 Phase Pickup Test							
Setting	Calculated Value	Phase A		Phase B		Phase C	
		Result	Error	Result	Error	Result	Error
4	20	20.2	1%	20.1	0.5%	20.2	FAIL
Comments:	PASS						
50 Timing Test							
Setting	Injected Current (A)	Phase A		Phase B		Phase C	
		Result(ms)	Pass or Fail	Result(ms)	Pass or Fail	Result(ms)	Pass or Fail
4	30	20.4	PASS	17.8	PASS	19	PASS
Comments:	PASS						

	Company: Energized Electrical	Work Order: Northern Alberta Institute of Technology
	Location: Edmonton, AB	Date: 12/8/25
	Tested by: Dardon.J, Grewal.Y	

4.4 Basler-951 Relay

Inspection		
	OK / N-OK	Comments
Cleanliness Of Relay	OK	No dust was accumulated on the relay
Physical Condition	OK	No visible signs of wear and tear
HMI Operation	OK	HMI was operational
LED Operation	OK	All LED were working correctly
Relay Serial Number	E00962769	
Test Equipment		
Manufacture	Manta	
Model	MTS-5100	
Serial Number	13599	
Calibration Date	April 30, 2017	

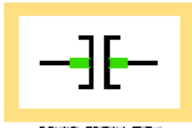
Setting								
Set Points		Value			Tolerance			
Pickup		2			2%			
Timing Curve		V2			N/A			
Time Dial		0.4			N/A			
51 Phase Pickup Test								
Setting	Calculated Value	Phase A		Phase B		Phase C		
		Result	Error	Result	Error	Result	Error	
2	2	2.001	0.1%	2.002	0.1%	1.997	0.2%	
Comments:	PASS							
51 Phase Timing Test								
Multiple of PU	Injected Current	Calculated Trip Time	Phase A		Phase B		Phase C	
			Result	Error	Result	Error	Result	Error
2x PU	4	0.76953	0.7076	0.1%	0.7021	0.1%	0.6954	0.1%
3x PU	6	0.207	0.3096	0.5%	0.3173	0.5%	0.3255	0.6%
4x PU	8	0.1027	0.2077	1%	0.1955	0.9%	0.2121	1.1%
Comments:	PASS							

	Company: Energized Electrical	Work Order: Northern Alberta Institute of Technology
	Location: Edmonton, AB	Date: 12/8/25
	Tested by: Dardon.J, Grewal.Y	

50 (Instantaneous Overcurrent) Phase Settings							
Set Points		Value			Tolerance		
Pickup		10			2		
Time Delay		0			<54.1ms		
50 Phase Pickup Test							
Setting	Calculated Value	Phase A		Phase B		Phase C	
		Result	Error	Result	Error	Result	Error
10	10.2	10	0%	10	0%	10	0%
Comments:	PASS						
50 Timing Test							
Setting	Injected Current (A)	Phase A		Phase B		Phase C	
		Result(ms)	Pass or Fail	Result(ms)	Pass or Fail	Result(ms)	Pass or Fail
10	15	24.2	PASS	22.9	PASS	29	PASS
Comments:	PASS						

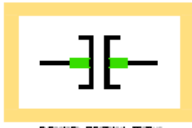
Directional Test

Inspection		
	OK / N-OK	Comments
Cleanliness Of Relay	OK	No dust was accumulated on the relay
Physical Condition	OK	No visible signs of wear and tear
HMI Operation	OK	HMI was operational
LED Operation	OK	All LED were working correctly
Relay Serial Number	E00962769	
Test Equipment		
Manufacture	MANTA	
Model	MTS – 5100	
Serial Number	13599	
Calibration Date	April 20,2017	

	Company: Energized Electrical	Work Order: Northern Alberta Institute of Technology
	Location: Edmonton, AB	Date: 12/8/25
	Tested by: Dardon.J, Grewal.Y	

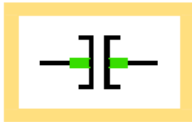
Metering Tests								
Voltage Magnitude & Angle Tests								
PT Ratio			120:120			PT Configuration		WYE
Injected Value			Calculated Reading			Actual Reading		
A	B	C	A	B	C	A	B	C
120	120	120	120	120	120	120.05	120.07	120.07
Current Magnitude & Angle Tests								
CT Ratio			200:5			CT Configuration		WYE
Injected Value			Calculated Reading			Actual Reading		
A	B	C	A	B	C	A	B	C
2	2	2	80	80	80	81.1	80.8	80.2

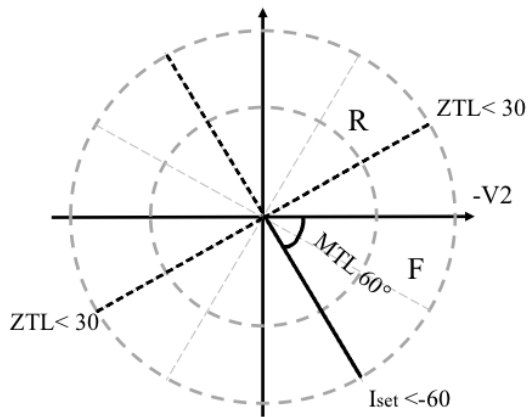
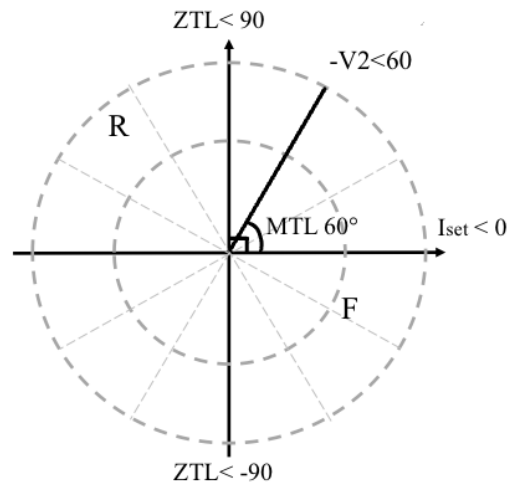
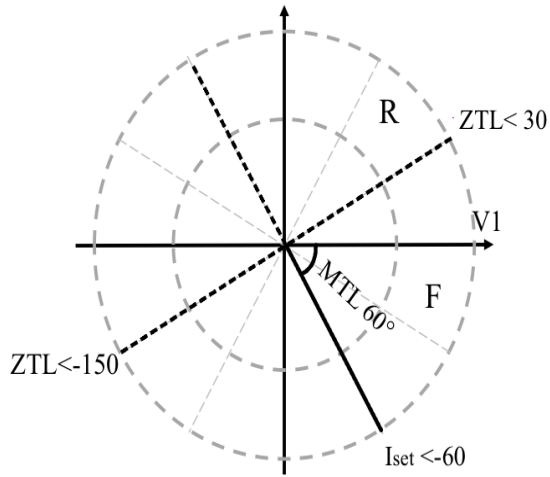
67 Directional Overcurrent Settings		
Setting	Value	Tolerance
Phase Element Pickup	3A	2% or 50mA
Phase Element Time Delay	0ms	
Positive Sequence Impedance Angle	60°	
Positive Sequence Impedance Magnitude	8 Ω	N/A
Neutral Element Pickup	1A	
Neutral Element Time Delay	0ms	
Zero Sequence Impedance Angle	60°	
Zero Sequence Impedance Magnitude	24Ω	

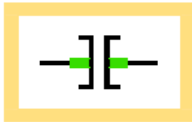
	Company: Energized Electrical	Work Order: Northern Alberta Institute of Technology
	Location: Edmonton, AB	Date: 12/8/25
	Tested by: Dardon.J, Grewal.Y	

67 Phase Test (Positive Sequence)					
Fault Voltages	Calculated Values	Test Results	Pick Up test (magnitude)	Lead Angle Test	Lag Angle Test
$V_a = 35$	$V_{pol} @ \angle 0$	I_A	2.996A	3.1A	3.1A
$V_b = 35$	$I_{set} @ \angle -60$	I_B	2.996A	3.1A	3.1A
$V_c = 35$	Lead angle @ $\angle 30$	I_C	2.996A	3.1A	3.1A
$V_{pol} = V_1$	Lag angle @ $\angle -150$	I_1	$\angle -60$	$\angle 29.61$	$\angle -150.59$
		Error	0.13%	1.3%	0.39%
67 Phase Test (Negative Sequence)					
Fault Voltages	Calculated Values	Test Results	Pick Up test (magnitude)	Lead Angle Test	Lag Angle Test
$V_a = 35$	$V_{pol} @ \angle 60$	I_A	2.996A	3.1A	3.1A
$V_b = 35$	$I_{set} @ \angle 0$	I_B	2.996A	3.1A	3.1A
$V_c = 69.27$	Lead angle @ $\angle 90$	I_C	0A	0A	0A
$V_{pol} = -V_2$	Lag angle @ $\angle -90$	I_{pol}	$\angle 0$	$\angle 89.71$	$\angle -90.40$
		Error	0.13%	0.32%	0.44%

67 Neutral Test (Zero Sequence)					
Fault Voltages	Calculated Values	Test Results	Pick Up test (magnitude)	Lead Angle Test	Lag Angle Test
$V_a = 35$	$V_{pol} @ \angle 0$	I_A	0.997A	1.1A	1.1A
$V_b = 69.27$	$I_{set} @ \angle -60$	I_B	0A	0A	0A
$V_c = 69.27$	Lead angle @ $\angle 30$	I_C	0A	0A	0A
$V_{pol} = -V_2$	Lag angle @ $\angle -150$	I_{pol}	$\angle -60$	$\angle 29.76$	$\angle -150.83$
		Error	0.3%	0.8%	0.55%

	Company: Energized Electrical	Work Order: Northern Alberta Institute of Technology
	Location: Edmonton, AB	Date: 12/8/25
	Tested by: Dardon.J, Grewal.Y	

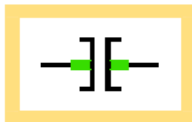


	Company: Energized Electrical	Work Order:
	Location: Edmonton, AB	Northern Alberta Institute of Technology
		Date: 12/8/25
Tested by: Dardon.J, Grewal.Y		

4.5 SEL-751 Relay

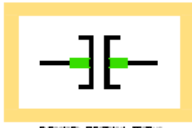
Inspection		
	OK / N-OK	Comments
Cleanliness Of Relay	ok	No dust was accumulated on the relay
Physical Condition	ok	No visible signs of wear and tear
HMI Operation	ok	HMI was operational
LED Operation	ok	All LED were working correctly
Relay Serial Number	3182710245	
Test Equipment		
Manufacture	Manta	
Model	MTS-5100	
Serial Number	13599	
Calibration Date	April 30, 2017	

Current Magnitude & Angle Tests								
CT Ratio			500:5			CT Configuration		
Injected Value			Calculated Reading			Actual Reading		
A	B	C	A	B	C	A	B	C
5	5	5	500	500	500	500.9	501.1	501.2

	Company: Energized Electrical		Work Order:
	Location: Edmonton, AB		Northern Alberta Institute of Technology
			Date: 12/8/25
	Tested by: Dardon.J, Grewal.Y		

51 (Time Overcurrent) Phase Settings								
Set Points		Value			Tolerance			
Pickup		2			0.02			
Timing Curve		U3			N/A			
Time Dial		1.5						
51 Phase Pickup Test								
Setting	Calculated Value	Phase A		Phase B		Phase C		
		Result	Error	Result	Error	Result	Error	
2	2	1.996	0.2%	1.997	0.15%	1.997	0.15%	
Comments:	Pass							
51 Phase Timing Test								
Multiple of PU	Injected Current	Calculated Trip Time	Phase A		Phase B		Phase C	
			Result	Error	Result	Error	Result	Error
2x PU	4	2.084	2.1	0.76	2.09	0.29	2.1	0.76
3x PU	6	0.872	0.884	1.02	0.881	1.02	0.885	1.99
4x PU	8	0.532	0.544	2.21	0.538	1.11	0.542	0.98
Comments:	Pass							

50 (Instantaneous Overcurrent) Phase Settings							
Set Points		Value			Tolerance		
Pickup		10			<1.5 cycles		
Time Delay		0					
50 Phase Pickup Test							
Setting	Calculated Value	Phase A		Phase B		Phase C	
		Result	Error	Result	Error	Result	Error
10	3%	10	0	10	0	10	0
Comments:	Pass						
50 Timing Test							
Setting	Injected Current (A)	Phase A		Phase B		Phase C	
		Result(ms)	Pass or Fail	Result(ms)	Pass or Fail	Result(ms)	Pass or Fail
10	15	20.1ms	PASS	24.5ms	PASS	22.2ms	PASS
Comments:	Pass						

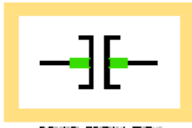
	Company: Energized Electrical	Work Order: Northern Alberta Institute of Technology
	Location: Edmonton, AB	Date: 12/8/25
	Tested by: Dardon.J, Grewal.Y	

4.6 SEL-587 Relay

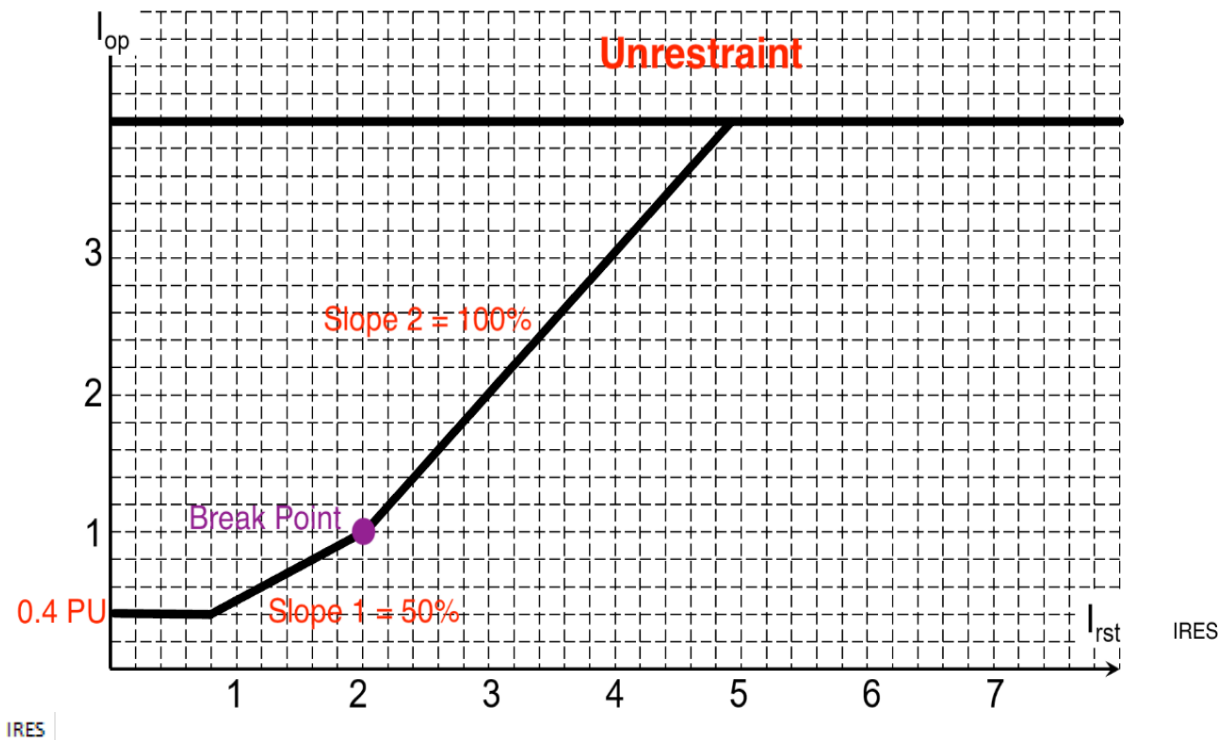
Inspection		
	OK / N-OK	Comments
Cleanliness Of Relay	ok	No dust was accumulated on the relay
Physical Condition	ok	No visible signs of wear and tear
HMI Operation	ok	HMI was operational
LED Operation	ok	All LED were working correctly
Relay Serial Number	2007100297	
Test Equipment		
Manufacture	Manta	
Model	MTS 5100	
Serial Number	13599	
Calibration Date	April 20,2017	

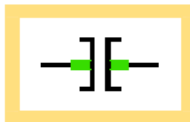
Metering Tests								
Current Magnitude & Angle Tests								
CT Ratio		1000:5		CT Configuration			YY	
Injected Value			Calculated Reading			Actual Reading		
A	B	C	A	B	C	A	B	C
5	5	5	1000	1000	1000	1001.3	1000.4	1001

87 (Differential) Settings							
MVA	VWDG1	CTR1	TR CON	SLP1	O87P	TAP1	PCT2
30	25	200	DABY	50	0.4	3.46	42
IRS1	VWDG2	CTR2	CT CON	SLP2	U87P	TAP2	PCT5
2	138	40	YY	100	4	3.14	30
87 Pick-up Test (Single Phase Test)							
Winding Pick-up Test		Setpoints & Calculations			Result		
		Setpoint	Tap	Pickup (A)	Pickup (A)	% Error	
W1 Minimum Pickup		0.4	3.46xA	1.384	1.368	1.17	
W2 Minimum Pickup		0.4	3.14xB	2.175	2.218	1.94	
W1 Unrestraint Pickup		4	3.46xA	13.84	13.772	0.49	
W2 Unrestraint Pickup		4	3.14XB	21.75	21.724	0.12	
Comments		A=1, B= $\sqrt{3}$					

	Company: Energized Electrical	Work Order: Northern Alberta Institute of Technology
	Location: Edmonton, AB	Date: 12/8/25
	Tested by: Dardon.J, Grewal.Y	

Harmonic Blocking Tests (Single Phase Test)						
Harmonic Blocking Test		Setpoint	W1 Test Current (A)	W1 blocking %	W2 Test Current (A)	W2 blocking %
2 nd Harmonic Blocking		42	2	43.06	4	42.03
5 th Harmonic Blocking		30	2	31.31	4	30.61
Comments:		Pass				
87 Slope Tests (Single Phase Test)						
	% Break Point	Calculated Value		Measured Result		
		I1 (A)	I2 (A)	I1 (A)	I2 (A)	
Slope 1	80	6.92	6.53	6.92	6.516	
	90	7.75	7.34	7.75	7.311	
Slope 2	110	9.688	8.701	9.688	8.708	
	120	10.726	9.246	10.726	9.234	
Comments:		Pass				

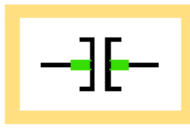


 ENERGIZED ELECTRICAL TESTING	Company: Energized Electrical	Work Order: Northern Alberta Institute of Technology
	Location: Edmonton, AB	Date: 12/8/25
	Tested by: Dardon.J, Grewal.Y	

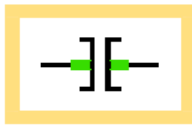
4.7 GE-469 Relay

General Data			
Relay Manufacturer	SEL	Hot / Cold Ratio	0.7
Relay Model	Sel-751	Motor FLA	300 A
Physical Condition	ok	Rated Voltage	2400 V
HMI Operation	ok	CT Configuration	4-W
Relay Serial Number	3182710245	PT Configuration	Open _delta
Relay Settings			
Phase CT Primary	500	U/C Alarm level	0.85 FLA / no delay
Phase CT Secondary	5	U/C Trip level	0.75 FLA
Ground CT Primary	50	U/C Trip delay	3 s
Ground CT Secondary	5	Mechanical Jam Trip	2 FLA
Overload Pickup	1.01 FLA	M/J delay	1 s
Overload Curve	2	U/B Alarm level	15% / no delay
If RTD Bias On	40°C, 130°C, 160°C	U/B Trip level	25%
Unbalance K factor	10	U/B Trip delay	2 s
RTD Alarm level	150°C	G.F Trip Level	0.4 CT
RTD Trip level	160°C	G.F Trip Delay	0 ms

Metering Tests								
Current Magnitude & Angle Tests								
CT Ratio		500:5		CT Configuration			Y	
Injected Value			Calculated Reading			Actual Reading		
A	B	C	A	B	C	A	B	C
1	1	1	100	100	100	99	99	99
Comments:		Pass						
Voltage Magnitude & Angle Tests								
PT Ratio		10:1		PT Configuration			Open DELTA	
Injected Value			Calculated Reading			Actual Reading		
A	B	C	A	B	C	A	B	C
10	10	10	173.2	173.2	173.2	171	174	173
Comments:		Pass						

	Company: Energized Electrical		Work Order: Northern Alberta Institute of Technology	
	Location: Edmonton, AB		Date: 12/8/25	
	Tested by: Dardon.J, Grewal.Y			

Testing				
RTD Tests	Settings	Result Temp.	Result delay (s)	Pass or Fail (Comments)
RTD Alarm	-----	-----	-----	
RTD Trip	160	162	0	Pass
Overload timing without RTD biasing	Settings	Injected I (A)	Result Time (s)	Pass or Fail (Comments)
150% O/L Timing	300	4.5	139	PASS
200% O/L Timing	300	6	58	PASS
RTD Biasing with Temp. Set at 130°C	Settings	Injected I (A)	Result Time (s)	Pass or Fail (Comments)
150% O/L Timing	300	4.5	95	PASS
200% O/L Timing	300	6	40	PASS
RTD Biasing with Temp. Set at 150°C	Settings	Injected I (A)	Result Time (s)	Pass or Fail (Comments)
150% O/L Timing	300	4.5	31	PASS
200% O/L Timing	300	6	13	PASS
Load Tests	Settings	Result condition	Result delay (s)	Pass or Fail (Comments)
U/B Alarm & Delay	15%/1s	31.36	2.20s	Pass
U/B Trip & Delay	25%/2s	46.05	2.067s	Pass
U/C Alarm & Delay	0.85xFLA/3s	2.50A	3.55	Failed
U/C Trip & Delay	0.75xFLA/2s	2.2A	3.21	Pass
Mech Jam Trip & Delay	2xFLA	6A	1.0751s	Pass
G.F Trip / Delay	0.4CT	2.016	0	Pass
Phase Reversal Trip	100%	100%	0.55s	Pass
Comments	U/C Alarm & Delay needs to be within 0.5s of setting			

	Company: Energized Electrical	Work Order: Northern Alberta Institute of Technology
	Location: Edmonton, AB	Date: 12/8/25
	Tested by: Dardon.J, Grewal.Y	



2. Deficiencies and Recommendations

Westinghouse CO relay

Through testing the Westinghouse C0-8 relay as per NETA MTS standards no deficiencies were found with testing the 50/51 and DC seal-in elements. All Testing passed the required Tolerances listed by the manufacturer. For the Mechanical testing, the Westinghouse flag mechanism for 50 and 51 functions is damaged.

Going forward, the Westinghouse relay would need a replacement for the damaged flag mechanism. basic maintenance needs to be done to maintain then to maintain the relay's longevity. Cleaning the relays and tightening wire connections quarterly and maintaining wire health will ensure the relays operation.

GE-735

Through testing the GE-735 relay, as per NETA MTS standards. The GE-735 relay only passed the 50/51 pick testing but failed the 51-timing test. Though testing the 51-timing element, a percent error was calculated far past the manufacturers' recommended data.

The timing tolerance should be no more than 20ms. The GE-735 relay didn't pass that threshold. More testing will need to be done to determine the cause of the timing element. Going forward, the GE-735 should be taken out of service till the cause of the 51-timing element is found.



GE-750

Through testing the GE-750 relay as per NETA MTS standards, no deficiencies were found, with testing the 50/51 elements. All Testing passed the required Tolerances listed by the manufacturer.

Going forward, the GE-750 relay would need some basic maintenance done to maintain the relays longevity. cleaning the relays and tightening wire connections quarterly and maintaining wire health will ensure the relays operation.

Basler-951

Through testing the Basler-951 relay as per NETA MTS standards no deficiencies were found, with testing the 50/51,67P and 67N elements. All Testing passed the required Tolerances listed by the manufacturer with low percent error.

Going forward, the Basler-951 will be able to stay in use for the far future. Some basic maintenance done to maintain the relay's longevity and accuracy. Clearing the relay area, tightening wire connections and ensuring the condition of wire connections quarterly will ensure the relays operation.

SEL-751

Through testing the SEL-751 relay as per NETA MTS standards no deficiencies were found, with testing the 50/51 elements. All Testing passed the required Tolerances listed by the manufacturer with low percent error.

Going forward, the SEL-751 will be able to stay in use for the far future. Some basic maintenance done to maintain the relay's longevity and accuracy. Clearing the relay area, tightening wire connections and ensuring the condition of wire connections quarterly will ensure the relay's operation.



SEL-587

Through testing the SEL-587 relay as per NETA MTS standards no visible deficiencies were found. For testing the restrained and Unrestrained current pickup operation, restraint slopes and harmonic blocking. All Testing passed the required Tolerances listed by the manufacturer with low percent error.

Going forward, the SEL-587 will be able to stay in use for the far future. Some basic maintenance will need to be done to maintain the relay's longevity and accuracy. Cleaning the relays and tightening wire connections quarterly and maintaining wire health by proacting proper wire handling will ensure the relays operation.

GE-469 Relay

Through testing the GE-469 relay as per NETA MTS standards no visible deficiencies were found. For testing the overload, Current Unbalance, Undercurrent, Mechanical Jam, phase reversal and Ground Fault elements. All elements passed except for the undercurrent alarm.

Through our testing, it was found that the alarm signal would go off within 0.55s of a trigger. To pass, it needed to go off within a maximum of 0.5s of the alarm trigger as per the manufacturer's data. Through our testing of the other elements, although they passed testing they are nearing allowable errors. Going Forward a replacement should be investigated in case of more future frailer in the relay.